MOBILE APPLICATION DEVELOPMENT LAB

Model Question Bank (Comprehensive Multi-Activity Problems)

Important Exam Note: These questions are curated strictly in accordance with the prescribed syllabus, evaluating core UI layout structure, runtime form validation, explicit/implicit intents, component communication, dynamic list rendering via Adapters, dialog popups, and state persistence using Shared Preferences. *(Note: All database/SQLite operations are omitted as per instructions.)*

Question 1: Patient Appointment & Clinic Dashboard

Design and develop an Android application for a **"Patient Appointment & Clinic Dashboard"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a patient's Full Name, Age, and Preferred Token/Date, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Patient's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the patient by retrieving their name from preferences and use an Android Adapter to render a collection of available specialist doctors inside a ListView. Clicking a doctor item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external web browser or mapping application for hospital route navigation.

Question 2: Smart Fitness Tracker & Workout Portal

Design and develop an Android application for a **"Smart Fitness Tracker & Workout Portal"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a user's Username, Target Weight, and Daily Calorie Goal, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Username using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the user by retrieving their name from preferences and use an Android Adapter to render a collection of fitness workout routines inside a ListView. Clicking a routine item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external web browser to stream instructional video tutorials.

Question 3: Student Course Enrollment & Academic Hub

Design and develop an Android application for a **"Student Course Enrollment & Academic Hub"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a student's Admission Number, Department Name, and Contact Number, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Admission Number using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the student by retrieving their identifier from preferences and use an Android Adapter to render a collection of elective courses inside a ListView. Clicking a course item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external browser to download the corresponding course syllabus document.

Question 4: Hotel Room Booking & Concierge Desk

Design and develop an Android application for a **"Hotel Room Booking & Concierge Desk"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a guest's Primary Name, Passport/ID Number, and Duration of Stay, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Guest's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the guest by retrieving their name from preferences and use an Android Adapter to render a collection of available hotel amenities and dining halls inside a ListView. Clicking an amenity item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to open a mapping application to pinpoint the exact floor or wing layout location.

Question 5: Corporate Job Application & Talent Portal

Design and develop an Android application for a **"Corporate Job Application & Talent Portal"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a candidate's Full Name, LinkedIn Profile URL, and Years of Experience, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Candidate's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the applicant by retrieving their name from preferences and use an Android Adapter to render a collection of open job openings inside a ListView. Clicking a job item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to open the phone dialer pre-loaded with the HR department's contact number.

Question 6: Vehicle Rental System & Drive Dashboard

Design and develop an Android application for a **"Vehicle Rental System & Drive Dashboard"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a driver's Full Name, Driving License Number, and Mobile Number, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Driver's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the driver by retrieving their name from preferences and use an Android Adapter to render a collection of available car and bike categories inside a ListView. Clicking a vehicle item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external web browser displaying terms of service and insurance policy rules.

Question 7: Culinary Recipe Finder & Chef Kitchen

Design and develop an Android application for a **"Culinary Recipe Finder & Chef Kitchen"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a chef's Profile Name, Dietary Preferences, and Allergy Information, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Profile Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the user by retrieving their name from preferences and use an Android Adapter to render a collection of gourmet cooking recipes inside a ListView. Clicking a recipe item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch a web browser mapping directly to video cooking guides.

Question 8: E-Commerce Product Launch & Pre-Order System

Design and develop an Android application for an **"E-Commerce Product Launch & Pre-Order System"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a customer's Full Name, Shipping Address, and Postal Code, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Customer's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the shopper by retrieving their name from preferences and use an Android Adapter to render a collection of premium limited-edition electronics inside a ListView. Clicking a product item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to open a web view showing comprehensive user manual documentation.

Question 9: Travel Itinerary Planner & Tourism Guide

Design and develop an Android application for a **"Travel Itinerary Planner & Tourism Guide"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a traveler's Passport Name, Destination Country, and Budget Limit, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Traveler's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the explorer by retrieving their name from preferences and use an Android Adapter to render a collection of top sightseeing landmarks inside a ListView. Clicking a landmark item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to load geographical map coordinates on an external application.

Question 10: Alumni Network & Placement Connect

Design and develop an Android application for an **"Alumni Network & Placement Connect"** by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing an alumnus's Full Name, Graduation Year, and Current Company, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Alumnus's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the user by retrieving their name from preferences and use an Android Adapter to render a collection of mentorship tracks and networking chapters inside a ListView. Clicking a chapter item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to open the device email application with the chapter lead's address pre-populated.